



ABSTRACT

This project presents an intelligent E-Learning platform for **BS-IT, University of the Punjab**. It is built on the **MERN Stack** with **Firebase Authentication** (Email & Google). The system offers 3 roles: **Student, Teacher, Admin**. Students can view courses, watch video lectures, submit assignments, and get instant help via **AI Chatbot** powered by **Gemini API**. Teachers can create courses and upload lectures. Admin manages users and courses. It provides a customized, secure, and scalable solution to replace generic platforms like Udemy for department-level learning.



OBJECTIVES

- ✓ **Centralized LMS** for 8 Semesters BSIT (PU)
- ✓ **Secure Authentication** via Firebase
- ✓ **Role-Based Dashboards** (Student, Teacher, Admin)
- ✓ **Video Lectures & Assignment Portal**
- ✓ **AI Chatbot** for 24/7 Student Support



Secure Login
(Email & Google)



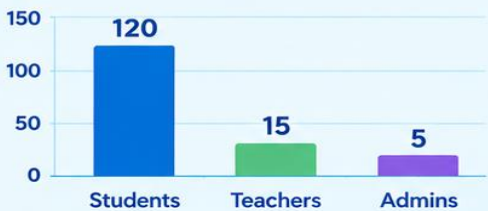
Multi-Role Access
(Student / Teacher / Admin)



Video Lectures
(Watch & Learn)



User Engagement



Course Completion Rate



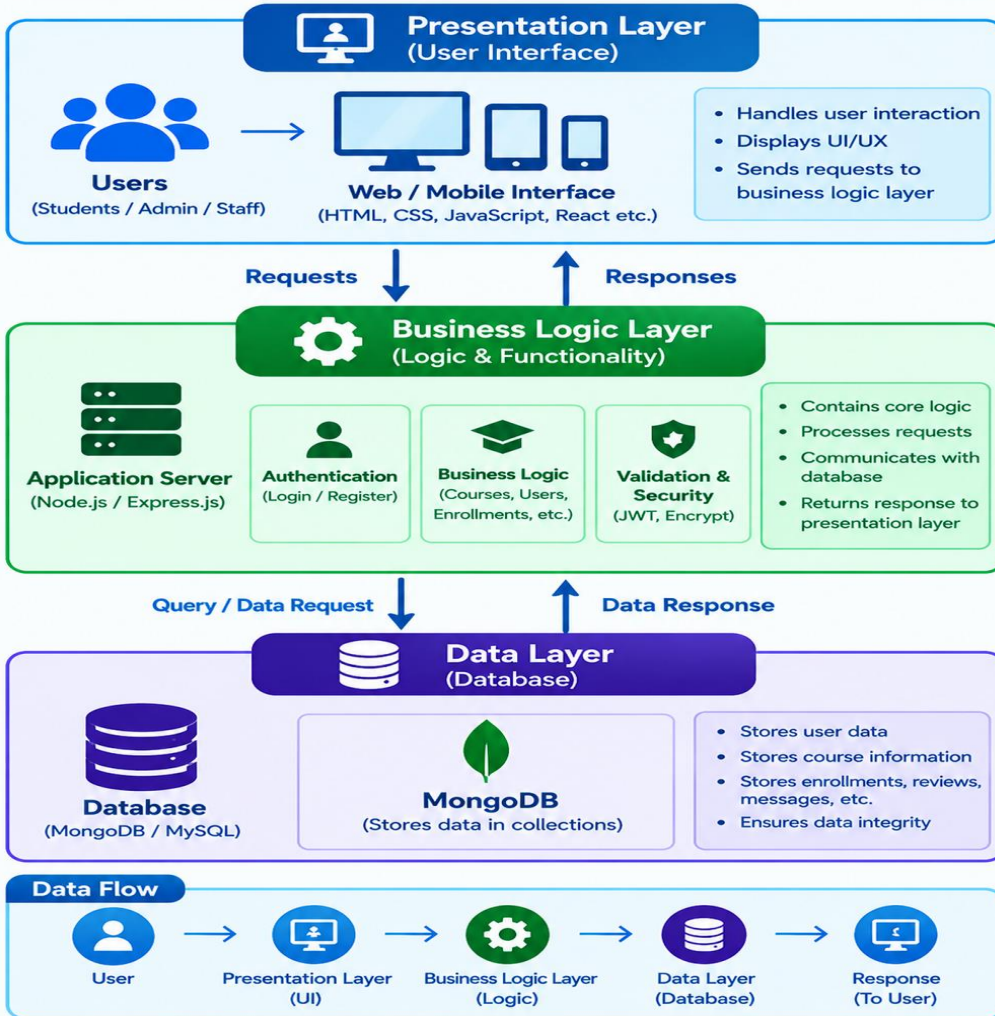
AI Chatbot Usage



System Performance



METHODOLOGY



RESULTS & DISCUSSIONS



CONCLUSION

The “Revolutionizing Digital Learning With AI” project successfully provides an online learning platform for students, teachers, and admin. Students can access courses, lectures, assignments, and an AI chatbot. Teachers can create courses, lectures, and assignments, while admins can manage courses and users.

The system was tested through different test cases, and all 17 test cases passed successfully. This shows that the main features of the system are working properly.

Overall, the project provides a simple and centralized way for students to learn and for teachers to manage educational content. In the future, features such as an AI Quiz Generator, Android and iOS applications, and offline access can be added.

